Chemical Separations for a Cleaner World: Coursework [30 marks in total]

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Please email by Monday 1700 to inform us if you will be coming to the Drop-In Session (if not, the session will be canceled)!

Submission Deadline: December 04th at 0000 (GMT)

Assessment Type: Group Coursework

Mission O_2: A Journey to the Ideal Adsorbent for Exploring Europa

You *(in this case your group)* are part of an elite team of process engineers working in an underground colony in Europa, Jupiter's moon whose oxygen (O_2) concentration in its nitrogen (N_2) atmosphere is low. You are on a mission to crack one of the most critical challenges in exploring the moon – creating a medical oxygen concentrator that delivers high-purity oxygen for the explorers. O_2, as you would know is critical for survival of the brave explorers during exploration and making sure it is readily available is essential. Your colony has embarked on an ambitious project to develop the best possible oxygen concentrators, and you have been tasked with choosing the perfect adsorbent material to separate oxygen from nitrogen (N_2) in Europa's atmosphere. In an oxygen concentrator, nitrogen-rich air flows through a packed column filled with an adsorbent. The adsorbent's role is to selectively remove nitrogen so that only oxygen is enriched at the outlet. Your team has narrowed down the potential adsorbents to two high-performing candidates. Now, it is up to your team to evaluate these materials and identify the best material that will finally facilitate the exploration of Europa.

To determine the best adsorbent, you will need to conduct two types of tests: static and dynamic measurements. Static measurements provide a snapshot of the material's nitrogen and oxygen adsorption capacity. Using a volumetric apparatus, your colleagues in the materials science team have already measured adsorption isotherms for two promising materials, *Romulus* and *Remus*, to determine how much N_2 and O_2 each adsorbent can adsorb. The measurements were conducted over a range of partial pressures at 303 K and the adsorption capacity is provided in Table 1. Dynamic measurements facilitate simulating real-world conditions where nitrogen-rich air flows continuously through the adsorbent bed. The breakthrough studies will allow you to measure how well each material performs by providing valuable insights on how long the adsorbent can adsorb nitrogen before reaching its capacity and contaminating the pure oxygen stream. However, before conducting the breakthrough studies, the Government of Europa is eager to establish a mathematical model for simulating these breakthrough experiments on a computer. Using the data from the static measurements, your model will be crucial in predicting adsorbent performance without exhaustive physical testing, saving time and resources in a hostile environment.

Prepare a 2-page technical note for the Government of Europa, outlining the process of selecting the optimal adsorbent for oxygen concentrators. The report should cover the conclusions drawn from static measurements, how these findings contributed to building an adsorption model, and how this model informed the design decisions in developing the oxygen concentrators.

Note: The problem is open-ended, allowing for a range of approaches. As long as you provide clear justifications and thorough scientific analysis grounded in the concepts covered in the module, your work will be assessed favorably.

Static measurements

The adsorption data for Romulus and Remus is in a discrete format, making it incompatible for direct use in the model you will develop for the breakthrough studies. Using the data provided in Table 1, identify an appropriate isotherm model, estimate the isotherm parameters, and evaluate the performance of the adsorbents by comparing their adsorption capacity and selectivity.

Table 1: Isotherm data for N_2 and O_2 on adsorbent *Romulus* and *Remus* at 303 K.

Romulus				**Remus**			
N_2		**O_2**		**N_2**		**O_2**	
P [Pa]	q [mol kg^{-1}]	P [Pa]	q [mol kg^{-1}]	P [Pa]	q [mol kg^{-1}]	P [Pa]	q [mol kg^{-1}]
0.000	0.000	0.000	0.000	0	0.000	0	0.000
2026.500	0.025	8106.000	0.015	63400	0.1674	33200	0.0372
3039.750	0.038	20265.000	0.035	88700	0.2300	61300	0.0679
4053.000	0.053	247435.650	0.366	117500	0.2994	96100	0.1022
6079.500	0.079	371153.475	0.521	162500	0.3934	143200	0.1504
8106.000	0.108	494871.300	0.693	208500	0.4848	214800	0.2138
61909.575	0.562	618690.450	0.815	303800	0.6450	355900	0.3455
111356.175	0.835			427000	0.8280	548200	0.5089
173265.750	1.082			609500	1.0588	747900	0.6830
247435.650	1.295			776600	1.2249	859000	0.7679
321706.875	1.482			919000	1.3472	931700	0.8348
395978.100	1.597						
470148.000	1.714						
544419.225	1.795						
618690.450	1.861						

Dynamic measurements

As outlined in the problem statement, your task is to develop a mathematical model to simulate dynamic breakthrough experiments. Use the isotherm models derived from static measurements for both *Romulus* and *Remus* to simulate the breakthrough of pure N_2 when the adsorption column (both solid and fluid phases) is initially saturated with pure O_2. Evaluate the performance of the adsorbents by comparing their breakthrough curves/times. For your reference, the column dimensions and gas physical properties are provided in Table 2.

This model will guide design decisions to maximize adsorbent performance, allowing oxygen production for extended periods before regeneration is needed. Conduct a study to inform the selection of optimal operating conditions (e.g. feed velocity) and column dimensions (e.g. column length) to maximize performance.

Table 2: Column dimensions, physical and operating conditions used in the simulations.

Parameter	**Value**
Column dimensions	
Column length L [m]	0.32
Column void fraction ϵ	0.32
Particle density of adsorbents A and B ρ_s [kg m^{-3}]	990
Particle radius r_p [m]	0.001
Properties and constants	
Tortuosity τ	3
Molecular diffusivity D_m [m^2 s^{-1}]	7.481×10^{-5}
Universal gas constant R [m^3 Pa mol^{-1} K^{-1}]	8.314
Operating conditions	
Process temperature T [K]	303
Process pressure P [kPa]	100
Inlet velocity u_{in} [m s^{-1}]	4.864×10^{-3}

Material Required for Submission

1. Please publish the code for the model (including parameter estimation) to a Git-hosting platform (e.g. GitHub). *You may use MATLAB, Python (strongly recommended for career prospects), or C++*

   Note: Git is required for submission. We recommend you maintain your Git folder regularly, as it will track your contribution to the code. In case of conflicts, going through the Git history is the easiest way to resolve them. Resources for using git depending on your code editor: MATLAB, Visual Studio Code

2. Please submit a technical note through Canvas **[Total - 30 marks]**

   Constraints: The technical note should be limited to two pages. Use a 10pt Sans Serif font (e.g., Arial, Helvetica). Only one figure may be included, with a maximum size of 7 in. × 5 in.

   Note: We recommend you to check out LaTeX, an online document editor. It offers significant advantages over Microsoft Word, especially for ease of use, writing equations, and formatting documents—handling all formatting automatically, so you don't have to adjust anything manually! (Remember the days of the Design Project and struggling with formatting?)

   Suggested structure for the technical note (use as a guideline):

   - *Introduction **[Total - 3 marks]***
     - *Overview of the study **[1.5 marks]***
     - *Importance of evaluating adsorbents **[1.5 marks]***
   - *Methodology **[Total - 6 marks]***
     - *Methodology to obtain isotherm models (details of parameter estimation algorithms) **[1 mark]***
     - *Methodology to develop and implement a mathematical model for dynamic adsorption studies (details of differential equation solvers) **[3 marks]***
     - *State the operating conditions you have used in your simulations **[1 mark]***
     - *All relevant assumptions must be clearly stated and justified **[1 mark]***
   - *Results **[Total - 15 marks]***
     Provide a discussion, supported by a figure, that addresses:
     - *Isotherm model, parameters, comparison between experimental data and the fitted model **[4 marks]***
     - *Discussion on the evolution of the adsorption process obtained from breakthrough simulations on the two adsorbents for a given operating condition and column dimension **[4 marks]***
     - *Discussion on the sensitivity analysis on operating conditions and column dimensions **[3 marks]***
     - *Comparison with equilibrium theory (also provides a way to check your modeling results) **[4 marks]***
   - *Limitation **[Total - 6 marks]***
     - *Discussion on the limitations of your model **[3 marks]***
     - *Avenues for improvement**[3 marks]***

3. Please submit a table of contribution

   On the first page of your submitted technical note, include the details of each group member (name, student ID, email address) along with their specific contributions to the coursework. Include a link to your code on this page. This page will not count toward the page limit, and only the requested information should be included. An example of a credit author statement is available here for your reference.

List of Equations

Component mass balance:

$$\frac{\partial y_i}{\partial t} = D_L \frac{\partial^2 y_i}{\partial z^2} - \frac{\partial}{\partial z}(y_i v) - \frac{RT}{P}\frac{\rho_s(1-\epsilon)}{\epsilon}\frac{\partial q_i}{\partial t}, \quad \sum y_i = 1 \tag{1}$$

$$D_L = 0.7 D_m + v_{\text{in}} r_p \tag{2}$$

Boundary conditions:

$$\text{For } z = 0: \quad y_i(z=0) = y_{i,\text{feed}} + \frac{D_L}{v} \left.\frac{\partial y_i}{\partial z}\right|_{z=0} \tag{3}$$

$$\text{For } z = L: \quad \left.\frac{\partial y_i}{\partial z}\right|_{z=L} = 0 \tag{4}$$

Solid phase mass balance:

$$\frac{\partial q_i}{\partial t} = k_i \left(q_i^* - q_i\right) \tag{5}$$

Further Reading

Reza, H., Aniruddha, M., Ricky, N., Arvind, R., Shamsuzzaman, F., & Mohammad, A. (2013). Multiobjective Optimization of a Four-Step Adsorption Process for Postcombustion CO_2 Capture Via Finite Volume Simulation.

Peer Assessment

Overview

The group coursework is marked out of **30**. Each group receives the same base "group mark" for the submitted work. Peer assessment is then used to adjust this base mark, ensuring that individual contributions are fairly recognized.

$$\text{Final Individual Mark} = \text{Group Mark} \times \text{Peer Factor}$$

Peer Assessment Criteria (3 Points)

Each student rates their peers (not themselves) on the following three dimensions:

1. **Contribution to Work:** Effort in completing allocated tasks; quality of the work produced.
2. **Engagement & Initiative:** Attendance and responsiveness; taking responsibility, proposing ideas, and supporting progress.
3. **Teamwork & Collaboration:** Communication with others; helping and cooperating to achieve group goals.

Scoring Scale (0–5 per criterion)

0 No contribution whatsoever (only if someone has literally done nothing).
1 Much less than the average contribution.
2 Below average contribution.
3 About average contribution.
4 Above average contribution.
5 Much greater than the average contribution.

Note: A score of 3 is the baseline for an "average" group member.

Normalization Process

1. Each student's scores are averaged across the three criteria and across all raters.
2. The group average score is calculated.
3. Each student's peer factor is computed as:

$$\text{Peer Factor} = \frac{\text{Student Average}}{\text{Group Average}}$$

4. Peer factors are capped to lie within the range 0.8 – 1.2.
5. Final individual marks are calculated as:

$$\text{Final Mark} = \text{Group Mark} \times \text{Peer Factor}$$

Safeguards & Moderation

- All ratings must be accompanied by brief written comments justifying the score.
- The supervisor may moderate peer factors if scores appear unfair or inconsistent.
- Git authorship may be used as evidence as part of supervisor moderation.
- Failure to submit peer assessments results in a –0.05 penalty to the student's factor.
- Maximum mark remains capped at **30/30**.